3.(b) //STACK USING LINKED LIST

```
#include<stdio.h>
#include<stdlib.h>
struct node {
   int data;
   struct node *next;
};
struct node *top=NULL,*temp,*newNode;

void push(int data)
{
        newNode = (struct node*)malloc(sizeof(struct node));
        newNode->data = data;
        newNode->next = NULL;

        if(top == NULL)
        {
                top = newNode;
        }else{
                newNode->next = top;
                top = newNode;
        }
        printf("element pushed:%d \n",data);
}

int pop()
{
   if (top == NULL) {
      printf("\nEMPTY STACK");
   } else {
      temp = top;
      int temp_data = top->data; //to store data of top node
      top = top->next;
      free(temp); //deleting the node
      return temp_data;
   }
}

void display() {
   // Display the elements of the stack
   if (top == NULL) {
      printf("\nStack Underflow\n");
   } else {
      printf("\n The stack is: \n");
      temp = top;
      while (temp != NULL) {
```

```
            printf("%d--->", temp->data);
            temp = temp->next;
        }
        printf("NULL\n\n");
    }
}

int main()
{
        push(10);
        push(20);
        push(30);
        display();
        printf("\nelement poped %d",pop());
        printf("\nelement poped %d",pop());
        display();
        return 0;
}
```

OUTPUT:

```
element pushed:10
element pushed:20
element pushed:30

 The stack is:
30--->20--->10--->NULL


element poped 30
element poped 20
 The stack is:
10--->NULL
```